

Clinical Study Summary Report (CSSR)

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1. Administrative & Study Identification

Full Study Title: A nationwide prospective patient-reported outcomes survey on the impact of COVID-19 **Short Title/Acronym:** COVID-19 PRO Study **Protocol Number:** C4591034 **NCT Number:** NCT05160636 **Sponsor Name:** Pfizer Inc. **Investigational Product:** BNT162b2 (Comirnaty) / COVID-19 Vaccine (Analyzed as an observational exposure) **Phase of Development:** N/A (Observational Study) **Medical Monitor:** Pfizer Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the impact of acute COVID-19 infection on Health-Related Quality of Life (HRQoL), Work Productivity and Activity Impairment (WPAI), and the longitudinal prevalence of acute and long COVID symptoms among symptomatic adult outpatients. **Secondary Objectives:** To assess the effects of prior COVID-19 vaccination (BNT162b2, including bivalent boosters) on reducing the prevalence and severity of long-term COVID-19 symptoms and improving patient-reported outcomes compared to unvaccinated or not up-to-date individuals.

2.2 Background & Rationale To address the gap in evidence regarding the real-world longitudinal burden of long COVID symptom clusters, HRQoL, and work productivity impacts among outpatients with mild-to-moderate COVID-19. Furthermore, robust data is required to understand the protective effects of the BNT162b2 vaccine in accelerating recovery, reducing long COVID prevalence, and restoring patients' functional status and daily activities post-infection.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational (Prospective longitudinal patient-reported outcomes survey) **Control Type:** Unvaccinated / Not up-to-date cohort (Comparative exposure) **Blinding:** Open-label / Unblinded **Randomization:** N/A (Observational)

3.2 Planned Enrollment & Duration Target \$N\$: ~1,405 participants (676 consented in the 2022 cohort; 729 consented in the 2023 cohort) **Number of Sites:** Nationwide across the United States (~5,000 CVS Pharmacy test sites) **Estimated Study Start:** 31-Jan-2022 **Estimated Primary Completion:** Late 2023 (following 6-month tracking of the 2023 cohort)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years. 2. Positive reverse transcription-polymerase chain reaction (RT-PCR) or rapid antigen test result for SARS-CoV-2 at a CVS Health test site. 3. Self-reporting at least one symptom suggestive of acute SARS-CoV-2 infection at the time of testing.

4.2 Exclusion Criteria 1. Asymptomatic at the time of testing. 2. Incomplete or unverified COVID-19 vaccination schedules (excluded from specific vaccine comparative analyses). 3. Inability or refusal to complete the longitudinal digital surveys (e.g., lack of internet access).

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: N/A. This is an observational study. The primary "exposure" evaluated is prior vaccination status with BNT162b2 (primary series and/or booster) vs. unvaccinated status. **Route of Administration:** N/A (Intramuscular for prior vaccine exposure) **Duration of Treatment:** 6 months of observational follow-up post-infection.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care for symptomatic COVID-19 outpatient management as directed by the patient's healthcare provider. **Prohibited:** N/A.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening/Baseline	Day 0	Informed Consent, Demographics, SARS-CoV-2 Testing, Pre-COVID HRQoL Recall
Acute Follow-up	Day 3 & Week 1	EQ-5D-5L, WPAI-GH, PROMIS Fatigue, Acute Symptom Assessment
Post-Acute Follow-up	Week 4, Month 3	CDC-based Long COVID Symptom Questionnaire, EQ-5D-5L, WPAI-GH
End of Study	Month 6	Final CDC-based Long COVID Questionnaire, EQ-5D-5L, WPAI-GH

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Change in Health-Related Quality of Life (HRQoL) and the prevalence/number of long COVID-19 symptoms at Week 4, Month 3, and Month 6 following acute infection. **Measurement Method:** Electronic Patient-Reported Outcomes (PROs) using the EQ-5D-5L (Utility Index and EQ-VAS) and a CDC-based 20-symptom questionnaire.

7.2 Secondary & Exploratory Endpoints 1. Change in Work Productivity and Activity Impairment (WPAI-GH) outcomes (including absenteeism, presenteeism, and overall activity impairment). 2. Association and efficacy of BNT162b2 vaccination status on reducing long COVID symptom clusters and accelerating functional recovery.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Conducted via self-reported symptom tracking on digital surveys. As an observational study of a real-world illness, routine clinical trial AE reporting for an Investigational Product does not apply. **Serious Adverse Events (SAEs):** N/A (Observational PRO study). **Data Monitoring Committee (DMC):** Internal safety and data review by the Sponsor and research partners.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All consented participants with positive tests and complete survey data (Evaluable PRO population). **Sample Size Justification:** Convenience sampling based on testing volumes at national retail pharmacies, sized sufficiently to power multivariable logistic regression and latent class analysis (LCA). **Handling of Missing Data:** Mixed Models for Repeated Measures (MMRM) were used to estimate magnitude and account for missing data over time; per guidelines, no imputation was made when scoring EQ-5D-5L UI and WPAI instruments.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and ethical guidelines for survey-based human subject research. **Monitoring Plan:** Centralized data monitoring of electronic survey completion rates and quality. **Audit Trail:** Electronic data capture via digital surveys with automated logic checks, data queries, and adjudication algorithms for unmatched entries.

11. Study Identifiers & Governance

Primary ID: C4591034 **Registry Number:** NCT05160636 **Sponsor/Lead Organization:** Pfizer Inc. **Study Title:** A Study to Look at the Health Outcomes of Patients With COVID-19 and Influenza **Official Regulatory Title:** A nationwide prospective patient-reported outcomes survey on the impact of COVID-19

12. Clinical Framework (COVID-19 Specific)

Primary Condition: COVID-19 / Post-COVID-19 Condition (Long COVID) **Diagnosis Threshold:** Positive RT-PCR or rapid antigen test for SARS-CoV-2 **Medical Methodology:** Patient-Reported Outcomes (PRO) longitudinal tracking **Optimization Target:** Symptom burden reduction and HRQoL/work productivity recovery

13. Study Procedures & Methodology

Management Pathway: Observational tracking of standard outpatient COVID-19 trajectory **Education Protocol (HCP):** N/A **Education Protocol (Patient):** N/A **Quality Audit Mechanism:** Automated tracking of digital survey responses and statistical anomaly checks.

14. Observation & Follow-Up Schedule

Total Duration: 6 Follow-up months **Onsite Visit Cadence:** Baseline testing only (CVS Pharmacy) **Remote Monitoring Frequency:** Day 3, Week 1, Week 4, Month 3, Month 6 (via digital PRO surveys) **Checklist Review Interval:** Ad-hoc based on completed survey waves

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]				Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]				